Sparse Representation-Based Classification of Attributed Graphs

Raphaël Duy Phong Vo
raphael.phong@gmail.com

***Abstract*—Graph classification consists of assigning one label (out of a few possible ones) to an often-times annotated network structure. Graph neural networks (GNN), which leverage gradient-based optimization to learn well-suited nodal embeddings, constitutes the best solution up to date. One difficulty which characterizes the state-of-the-art message passing framework of GNNs is the graph readout step, where one attempts to summarize the global information of an attributed graph. Here, the risk of losing local information runs high. Returning to the more fundamental field of graph signal processing, this paper investigates the use of learnable dictionaries as a possible remedy to this problem. This new method of classification, called "Graph Sparse-Representation based Classification" (GraphSRC), shows all but competitive performance on three common benchmark datasets. New research directions are thusly proposed.**

***Index Terms*—Dictionary learning, graph classification, graph representation learning, graph signal processing, sparse coding**

I. Introduction

The graph continues to enjoy both economical and mathematical interest as the simplest topological structure over which signals can be defined, manipulated, and – hopefully – interpreted. Of great interest and yet of notorious difficulty is the fairly new task of classifying graphs endowed with an informative, node-wise signal (from here on referred to as "attributed graphs"). Tools previously developed by the signal processing community like spectral filtering [2], [3], wavelet transforms [1], sampling algorithms [4] and so on allow for a rigorous treatment of nodal signals, albeit in the limited context where only one graph is considered. Developped adjacently, graph neural networks – conceived from the paradigms of parameterized message propagation [5], [8] and learned filtering [6] – offer a heavy-duty solution for attributed graph classification within the setting of multiple graphs. Still, one notable challenge of this task lies in the "reading out" of each graph. It is not evident to devise a uniform procedure that takes an attributed graph and returns a summary of its global information, because the number of nodes (i.e., the signal length) and the domain topology varies from graph to graph. One popular remedy is the graph pooling framework [7], which essentially performs recursive subsampling on the graph (often interlaced with the message propagation/filter layers) followed by the application of a global, permutation-invariant function ϕ to each graph subsample (e.g. a signal mean across the remaining nodes); the results are combined with some operation $\oplus$ (e.g. summation or concatenation) to give the graph readout [8]. However, a common issue lies in the summative nature of ϕ, whereby loss of local information is unavoidable. Moreover, the pooling operation often requires an intricate reconstructive step where new edges need to be assigned between the remaining nodes after subsampling, hence requiring a supplementary learnable pair-wise operator to be iterated over all pairs of nodes.

In light of these inconveniences, we revisit the approaches of sparse representation-based classification [10], employing a bank of spectral kernels to sparsely encode signals across graphs with the hopes of providing an alternative to pooling for graph classification. After having bridged the gap between the traditional time-domain and the nodal domain with regards to dictionary learning theory in Sections III and IV, we provide in Section V a model which uses the same discriminative penalties as those of [12]. Data processing, training and experiment details are presented in Sections VI and VII.

II. Related work

A. Dictionary Learning for Classification

First, an early approach to the problem is to define one dictionary for each signal class, in the hopes that the dictionary yielding the best reconstruction of a signal f corresponds to the true label of f. Such works include [10], [12], [13]; [12] adds a softmax penalty to endow the dictionaries with discriminative power.

Another approach is to apply a linear classifier to the sparse coefficients [14], [18]. Joint optimization of both the classifier and dictionaries are proposed in [15]. Ref. [16] moreover adds a label consistent penalty term which enforces an association between dictionary atoms and particular labels. Other methods of discrimination within this framework include the use of Fisher discrimination within the representation coefficients [17], [18]. Ref. [19] separately learns one sub-dictionary for each class, removes the redundant atoms shared between them, and then combines them all to form one dictionary. Inspired by the latter, [11] continues to enforce an atom-to-class association while distinguishing discriminating atoms from common atoms.

B. Dictionary Learning on Graphs

There already exists literature which extends the framework of dictionary learning to the case of graph signals, albeit without a focus on signal classification. To wit, Zhang et al. [20] devise a dictionary composed of multiple sub-dictionaries, each obtained by the modulation of the graph Laplacian's eigenvalues – this approach does not lend itself to the setting where signals may lie on different graphs. On the other hand, Thanou et al. [21] use learnable polynomials of the Laplacian to construct the sub-dictionaries. This approach is amenable to the "multiple graph" setting because dictionaries are constructed from kernels defined on the frequency domain common to all graphs. However, the authors impose a set of constraints (e.g. frame bounds [22] and smoothness) which ensure, in general, a good reconstruction quality. Since we really seek to discriminate across signal labels using the reconstruction errors, consistently good dictionaries are undesirable and so we forego these additional constraints.

It is important to note that the above dictionaries are all "convolutional" in nature [23]: since they are assembled from translated kernels, the atoms enjoy more localization, although incurring higher computational cost and possibly adding redundancy in the representation atoms. We instead propose to use the untranslated kernel, at the cost of having to diagonalize each graph Laplacian as a pre-processing step.

III. Preliminaries

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a graph with nodes $\mathcal{V}$ (say, N of them) and edges $\mathcal{E}$; also, we keep an implicit ordering on the nodes. If $\mathcal{A}$ and D are respectively the adjacency and the degree matrix of $\mathcal{G}$, then the normalized Laplacian of $\mathcal{G}$ is given by $\mathcal{L} = I - D^{-1/2}\mathcal{A}D^{-1/2}$. The Laplacian is positive-semidefinite, and as such always admits an eigendecomposition $\mathcal{L} = U\Lambda U^T$, with Fourier modes $U = [\chi_0, ..., \chi_{N-1}]$ and frequencies $\Lambda = \text{diag}(\lambda_0, ..., \lambda_{N-1})$. It follows that the graph Fourier transform of some $f : \mathcal{V} \to \mathbb{R}$ is given by $\hat{f} = U^T f$, while the inverse of $\hat{f}$ is given by $U\hat{f} = f$. Given a kernel $g : [\lambda_{\text{min}}, \lambda_{\text{max}}] \to \mathbb{R}$ over the spectrum, its nodal Fourier inverse is given by $\psi_g = U[g(\lambda_0), ..., g(\lambda_{N-1})]^T$, and its translation by node $i \in \mathcal{V}$ is given by a (normalized) convolution with the Dirac delta δ_i:

$$\psi_{g,i} := \sqrt{N} U g(\Lambda) U^T \delta_i.$$

As mentionned in [24], this definition of a convolution works backward from the notion that the graph Fourier transform should diagonalize the "node-domain" convolution $(\psi_g * \delta_i)$, but it is indeed well defined ([24], Section 5.3). More generally, $\sqrt{N} U g(\Lambda) U^T$ is the matrix $\sqrt{N} g(\mathcal{L})$, and the columns correspond to individual translates of the graph kernel. If g is a polynomial, then an intrusive diagonalization is not needed ([1], Section 6)

Thanou et al. build their dictionary by assembling the combined matrix $D = [U g_1(\Lambda) U^T, ..., U g_M(\Lambda) U^T]$. In general, when g is a smooth kernel with uniformly bounded derivatives, the magnitude of $\psi_{g,i}(n)$ will enjoy a quantifiable decay as the hop-distance between nodes i and n increases ([24], Corollary 2). This localization is desirable when one wants to perform a windowed convolution on some signal f with the kernel g, and may contribute positively to a better signal reconstruction. However, as previously mentioned, we want finer control over the dictionary where, ideally, each atom has its own set of parameters. The goal is to allow for an easier enforcement of discriminative capabilities, which is prioritized over reconstructive capabilities. To this end, we construct our dictionaries with the untranslated kernels ψ_g.

IV. Dictionary learning on graph signals

A. Block-matrix formulation

In practice, graphs may support signals with C different channels, as is the case when each node is assigned a vectorial embedding [25]. The graph signal is then actually a matrix $F = [f_1, ..., f_C] \in \mathbb{R}^{N \times C}$. The goal, then, is to construct a learnable dictionary composed of C-channel atoms, say M of them. To this end, we outline an approach highly similar to Thanou et al., but the construction of our dictionary is different in that we assemble a block matrix of the form

$$D = \begin{bmatrix} \psi_{g_{1,1}} & \cdots & \psi_{g_{1,M}} \\ \vdots & \ddots & \vdots \\ \psi_{g_{C,1}} & \cdots & \psi_{g_{C,M}} \end{bmatrix}. \tag{1}$$

The graph signal to be reconstructed then corresponds to the flattened node embeddings $f = [f_1^T, ..., f_C^T]^T$. Finally, one optimizes over $\mathbb{R}^M$ in search of the best atomic representation r^* such that

$$\min_{r \in \mathbb{R}^M} \frac{1}{2} ||f - Dr||_2^2 + \lambda ||r||_1, \tag{2}$$

is attained. Here, $\lambda > 0$ controls the enforcement of sparsity. One can also optimize over the kernel bank when the kernels have explicit parameterization, e.g. when they are polynomials. Consider the joint problem

$$\min_{\substack{r \in \mathbb{R}^M \\ \alpha_{1,1}, ..., \alpha_{C,M} \in \mathbb{R}^K}} \frac{1}{2} ||f - Dr||_2^2 + \lambda ||r||_1, \tag{3}$$

where $\alpha_{i,j}$ is a vector containing K polynomial coefficients such that $g_{i,j}(x) = \sum_{0 \leq k \leq K} \alpha_{i,j}(k) x^k$.

B. QCQP reformulation

Problem (3) is convex in $A = (\alpha_{1,1}, ..., \alpha_{C,M})$ when r is fixed, and vice-versa. For clarity of exposition, we derive a reformulation of Problem (3) which details the nature of A as an optimization variable. First, split the objective along the channels to get

$$\min_{A \in \mathbb{R}^{K \times CM}} \frac{1}{2} \sum_{1 \leq i \leq C} ||f_i - [\psi_{g_{i,1}}, ..., \psi_{g_{i,M}}] r||_2^2. \tag{4}$$

Now, define the vector γ^k by $\gamma^k = [\lambda_0^k, ..., \lambda_{N-1}^k]^T$ and let $\Gamma_K = [\gamma^0, ..., \gamma^K]$. Observe that

$$\psi_{g_{i,j}} = U g_{i,j}(\gamma^1) = U \left(\sum_{0 \leq k \leq K} \alpha_{i,j}^{(k)} \gamma^k \right) = U \Gamma_K \alpha_{i,j}.$$

In other words, letting $\alpha_i = [\alpha_{i,1}^T, ..., \alpha_{i,M}^T]^T$, the objective can be rewritten as a sum of independent quadratic programs:

$$\min_{\alpha_1, ..., \alpha_C \in \mathbb{R}^{KM}} \frac{1}{2} \sum_{1 \leq i \leq C} ||f_i - [r(1) U \Gamma_K, ..., r(M) U \Gamma_K] \alpha_i||_2^2 \tag{5}$$

The convexity of Problem (5) is useful when fitting A over the entire training set at once is feasible. A closed form update rule can even be derived by zeroing the Sample Average Approximated (SAA) gradient [26]. Still, the overall optimizing task is non-convex w.r.t. (A, r) and so we would still need to employ an alternating optimization scheme going back and forth between A and r [27].

V. Graph classification model

A. Stochastic gradient descent model

Often times, the dataset consists of relatively bare graphs requiring a more informative nodal signal (for instance, the signal might initially only consist of one-hot encodings). In this case, a node embedding model with parameters φ is used to provide each graph with a high-dimensional signal as the first part of the forward step. Only then does the sparse coding and classification occur, after which the parameter update takes place, now including an update of the embedder model's parameters. We use the Graph Isomorphism Network with skip connection as our node embedding model [30]. To accomodate the update of φ, the overall parameter optimization is best done through stochastic gradient descent. However, problem (5) is solved with a preconditioned conjugate gradient descent method to provide a warm start for our kernel coefficients (see Appendix A).

Let f belong to class i, $1 \leq i \leq L$. We decide to implement a classification model which relies on the reconstruction error of L distinct dictionaries, one for each class. Discrimination is enforced through the softmax penalty of Mairal et al. [12] defined by

$$C_i^T(y_1, ..., y_L) = \log\Big(\sum_{1\leq j\leq L} e^{-T(y_j - y_i)}\Big)$$

as originally presented, where T is a temperature parameter. Our corresponding problem statement regarding f is

$$\min_{\varphi, D_1, ..., D_L} C_i^T(y_1, ..., y_L) + T\gamma y_i$$

where each y_i is the reconstruction error of each of the L dictionaries regarding f computed in the sparse coding step as outlined by (2). For simplicity, assume that each dictionary D_j consists of p atoms.

B. Classification

Sparse coding coefficients r^* are computed by minimising $O(D, r)$ with the Fast Iterative Shrinkage-Thresholding Algorithm (FISTA) [28]. The final task of classification given r^* can be made in a number of ways. Ref. [11] achieve best results with a classifier which picks the label

$$\arg\min_i \{||f - D_i r_i^*||_2^2 + \lambda ||r_i^*||_2^2\},$$

where r_i^* is the subvector $[r^*(ip), ..., r^*((i+1)p-1)]$. Whether the sparse coding is done simultaneously over the whole dictionary (global classification) or individually over the sub-dictionaries (local classification) is a data-dependent decision. For abbreviation purposes, we call this method of graph labelling the method of "Graph Sparse-Representation Based Classification" (GraphSRC).

VI. Model optimization

A. Parameter updates

When graph signals are provided by a learnable node embedding parameter, the dictionary kernel coefficients are updated through mini-batched stochastic gradient descent on (6) with Adaptive Moment Estimation (Adam) [29]. One backward step consists of an update of the node embedding parameters φ and the kernel coefficients $A = (\alpha_{1,1}, ..., \alpha_{C,M})$.

VII. Experiments

A. Datasets

Three popular classification benchmarks, all of which (MUTAG, ENZYMES, PROTEIN) stem from bioinformatics, are used to assess the performance of our model. The ENZYMES dataset models 600 tertiary protein structures, each categorised under one of the top-level Enzyme Commission classes. PROTEINS consists of 1,113 proteins that are all either enzyme or non-enzyme. MUTAG contains 188 nitro-aromatic compounds which may or may not exhibit mutagenicity.

B. Experiment and Training Parameters

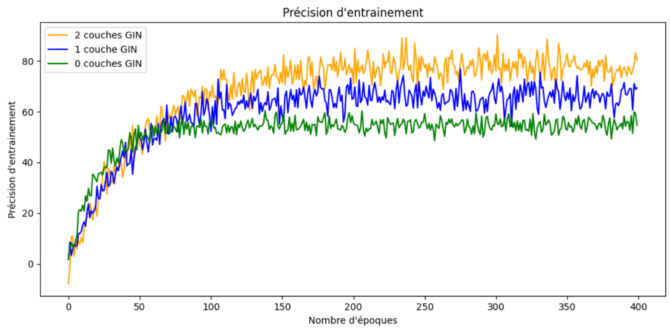
Following the work of [31], we put aside a hold out test set comprising 10% of the dataset and split the remaining portion into 10 folds for cross-validation. Validation accuracy is assessed every 5 epochs for each fold. The epoch which yields the best validation accuracy averaged across the 10 folds is recorded and used to re-train the final model on the entire non-test portion. An initial learning rate of 0.01 is used with a decay rate of 0.5 every 50 epochs. The temperature parameter T is searched within {0.1, 0.5, 1.0} and the fidelity parameter γ is searched within {0.001, 0.01, 0.1}, i.e. the pair (T, γ) is searched across the resulting 3x3 grid. As opposed to [12], we do not let the parameters T, γ vary. The number of atoms per class is restricted to {10,20,30} and the sparsity control is searched within {0.001, 0.01, 0.1} in a grid-search manner as well.

For the remaining hyperparameters, no grid search was conducted and instead a common basic model with at most one varying hyperparameter was used for the search. The number of coefficients K is searched throughout {4,6,8,10,12,14,16}.

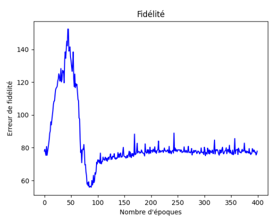
C. Performance and Ablation Studies

Using training accuracy as a proxy, the results of our experiment suggest that using learnable node embeddings increases model expressivity (Figure 1). Test set accuracies in Table 1 similarly report better generalization capabilities as we use more GIN layers.

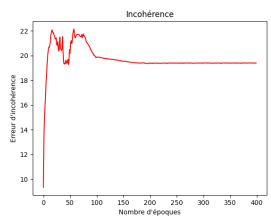
Increasing the number of atoms per class leads to overfitting and, counter-intuitively, worse reconstruction fidelity down the line. When class-specific dictionaries are greater in size,



(a) Epoch-averaged training accuracies for GraphSRC with 0,1, and 2 GIN layers.



(b) Epoch-averaged fidelity error for GraphSRC with 2 GIN layers.



(c) Epoch-averaged incoherence error for GraphSRC with 2 GIN layers.

Fig. 1: Statistics of the training procedure across 400 epochs.

TABLE I: Test set accuracies resulting from 10-fold cross-validation for GraphSRC with 0,1, and 2 GIN layers.

Model	PROTEINS	MUTAG	ENZYMES
GraphSRC (# GIN=0)	65.0 ± 3.3 %	71.3 ± 2.1 %	43.2 ± 3.9 %
GraphSRC (# GIN=1)	68.1 ± 2.8 %	75.4 ± 3.2 %	52.6 ± 4.6 %
GraphSRC (# GIN=2)	71.6 ± 2.2 %	79.2 ± 3.5 %	55.7 ± 3.1 %

between-class coherence of atoms explodes. This suggests that the gradient update scheme is ill-suited to increase discriminative power when there are too many atoms. We've observed that roughly 10 atoms per classes consistently outperforms the setting where 20 and 30 atoms per class is used; this is reflected in the stabilization of fidelity and incoherence in Figures 1(a) and 1(b). Unfortunately, the reconstruction error still remains undesirably high. Furthermore, even when the sparsity parameter is set high enough (so that around half of the dictionary coefficients are zero), the reconstruction error explodes when we use a higher number of atoms. Since we only ever use a low number of atoms, the sparsity parameter is best left at 0.001 as to not hinder signal fidelity.

Because we already employ a learning rate scheduling scheme where the gradient size purposefully decays every 50 epochs, increasing the temperature T of the softmax penalty is not an option because it would cancel out the gradient decrease. The only other choice closer to the approach of [12] is to decrease the fidelity parameter γ over time. This also engenders overfitting and reconstruction error however. Fixing γ at the same magnitude as the temperature ensures that reconstruction always remain a priority and helps attain a stable reconstruction error. This sensitivity to the training hyperparameters may be mitigated were we to pick a more stable optimizing scheme. Currently, the dictionaries are updated immediately following a change in the parameters of the node embedding model. There is no a priori knowledge of the distribution of the signals which we wish to simultaneously reconstruct and distinguish, making this dictionary learning task especially hard and prone to divergence.

The GIN should be configured as to output a small amount of channels, or else the sparse coding step becomes prohibitively expensive. We opt for a wide node-wise multi-layer perceptron with three output channels at each layer. With two layers, this culminates in a six channel output upon concatenation of the first layer and second layer output. Results show that the concatenation of two signals which each result from different scales of message passing allows for more expressivity and as such easier discrimination.

VIII. Conclusion

The compressed reconstruction of graph signals (through means of inverted Fourier kernels) essentially constitutes a new classification layer. As it stands, care must be taken in designing the training regime due to the inherent instability of the gradient update scheme. This problem may be mitigated by separately optimizing the embedding model and the dictionary (freeze one, train the other, and so on). Alternatively, a fixed embedding model could be designed with care to maximize only expressivity, ensuring that the training procedure be dedicated solely to optimizing the kernel coefficients. In both cases, we would be allowed to leverage the convexity of a full-batch dictionary update.

References

[1] D. K. Hammond, P. Vandergheynst, and R. Gribonval. “Wavelets on Graphs via Spectral Graph Theory”. *Applied and Computational Harmonic Analysis*, vol. 30, no. 2, pp. 129-150, 2011.

[2] A. Ortega, P. Frossard, J. Kovačević, J. M. F. Moura and P. Vandergheynst, ”Graph Signal Processing: Overview, Challenges, and Applications,” *Proceedings of the IEEE*, vol. 106, no. 5, pp. 808-828, May 2018.

[3] A. Gavili and X. -P. Zhang, ”On the Shift Operator, Graph Frequency, and Optimal Filtering in Graph Signal Processing,” *IEEE Transactions on Signal Processing*, vol. 65, no. 23, pp. 6303-6318, 1 Dec.1, 2017.

[4] Y. Tanaka, Y. C. Eldar, A. Ortega and G. Cheung, ”Sampling Signals on Graphs: From Theory to Applications,” *IEEE Signal Processing Magazine*, vol. 37, no. 6, pp. 14-30, Nov. 2020.

[5] Z. Wu, S. Pan, F. Chen, G. Long, C. Zhang and P. S. Yu, ”A Comprehensive Survey on Graph Neural Networks,” *IEEE Transactions on Neural Networks and Learning Systems*, vol. 32, no. 1, pp. 4-24, Jan. 2021.

[6] T. N. Kipf and M. Welling, “Semi-Supervised Classification with Graph Convolutional Networks,” in *ICLR*, 2017, pp. 1-14.

[7] L, Chuang, Y. Zhan, J. Wu, C. Li, B. Du, W. Hu, T. Liu, and D. Tao, ”Graph pooling for graph neural networks: Progress, challenges, and opportunities.” in *IJCAI*, 2023, pp.1-11.

[8] M. M. Bronstein, J. Bruna, T. Cohen, P. Veličković, “Geometric Deep Learning Models” in *Geometric Deep Learning: Grids, Groups, Graphs, Geodesics, and Gauges*, 1st ed.(TODO: May 4, 2021 by arxiv), pp. 77-80.

[9] Z. Zhang et al., ”Hierarchical Multi-View Graph Pooling With Structure Learning,” *IEEE Transactions on Knowledge and Data Engineering*, vol. 35, no. 1, pp. 545-559, 1 Jan. 2023.

[10] J. Wright, A. Y. Yang, A. Ganesh, S. S. Sastry and Y. Ma, "Robust Face Recognition via Sparse Representation," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 31, no. 2, pp. 210-227, Feb. 2009.

[11] S. Kong and D. Wang, "A Dictionary Learning Approach for Classification: Separating the Particularity and the Commonality," in *Computer Vision - ECCV 2012*, Berlin, Heidelberg, Germany: Springer Berlin Heidelberg, 2012, pp. 186-199.

[12] J. Mairal, F. Bach, J. Ponce, G. Sapiro and A. Zisserman, "Discriminative learned dictionaries for local image analysis," *2008 IEEE Conference on Computer Vision and Pattern Recognition*, Anchorage, AK, USA, 2008, pp. 1-8.

[13] K. Skretting, and J. H. Husøy. "Texture classification using sparse frame-based representations." EURASIP Journal on Advances in Signal Processing 2006, no. 1

[14] R. Grosse, R. Raina, H. Kwong, and A. Ng, "Shift-Invariant Sparse Coding for Audio Classification," in *UAI,* 2007, pp. 149-158.

[15] J. Mairal, J. Ponce, G. Sapiro, A. Zisserman, and F. Bach, "Supervised dictionary learning," in *Advances in neural information processing systems 21*, 2008.

[16] Z. Jiang, Z. Lin and L. S. Davis, "Label Consistent K-SVD: Learning a Discriminative Dictionary for Recognition," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 35, no. 11, pp. 2651-2664, Nov. 2013.

[17] M. Yang, L. Zhang, X. Feng and D. Zhang, "Fisher Discrimination Dictionary Learning for sparse representation," in *2011 International Conference on Computer Vision*, Barcelona, Spain, 2011, pp. 543-550.

[18] K. Huang and S. Aviyente, "Sparse Representation for Signal Classification," in *Neural Information Processing Systems*, 2006.

[19] I. Ramirez, P. Sprechmann, and G. Sapiro, "Classification and clustering via dictionary learning with structured incoherence and shared features," in *CVPR*, 2010.

[20] X. Zhang, X. Dong and P. Frossard, "Learning of structured graph dictionaries," in *2012 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, Kyoto, Japan, 2012, pp. 3373-3376.

[21] D. Thanou, D. I. Shuman and P. Frossard, "Learning Parametric Dictionaries for Signals on Graphs," *IEEE Transactions on Signal Processing*, vol. 62, no. 15, pp. 3849-3862, Aug.1, 2014.

[22] O. Christensen, "A short introduction to frames, Gabor systems, and wavelet systems," *Azerbaijan Journal of Mathematics*, pp. 25-39, 2014.

[23] C. Garcia-Cardona and B. Wohlberg, "Convolutional Dictionary Learning: A Comparative Review and New Algorithms," *IEEE Transactions on Computational Imaging*, vol. 4, no. 3, pp. 366-381, Sept. 2018.

[24] D. I. Shuman, B. Ricaud, and P. Vandergheynst, "Vertex-frequency analysis on graphs," *Applied and Computational Harmonic Analysis*, vol. 40, no. 2, pp. 260-291, 2016.

[25] W. L. Hamilton, R. Ying, and J. Leskovec, "Representation Learning on Graphs: Methods and Applications," *IEEE Data Engineering Bulletin*, vol. 40, pp. 52-74, 2017.

[26] S. Kim, R. Pasupathy, and S. G. Henderson, "A Guide to Sample Average Approximation," in *Handbook of Simulation Optimization*, M. C. Fu. Editor, Ed. 127, Springer, 2015, pp. 207-243.

[27] N. S. Chatterji and P. L. Barnett, "Alternating minimization for dictionary learning with random initialization," in *31st Conference on Neural Information Processing Systems (NIPS 2017*, 2017.

[28] A. Beck and M. Teboulle, "A fast Iterative Shrinkage-Thresholding Algorithm with application to wavelet-based image deblurring," in *2009 IEEE International Conference on Acoustics, Speech and Signal Processing*, Taipei, Taiwan, 2009, pp. 693-696.

[29] D. P. Kingma and J. Ba, "Adam: a Method for Stochastic Optimization," in *3rd International Conference on Learning Representations (ICLR 2015)*, 2015.

[30] K. Xu, W. Hu, J. Leskovec, and S. Jegelka, "How powerful are graph neural networks?" in Proc. Int. Conf. Learn. Representations (ICLR), 2019.

[31] C. Vincent-Cuaz, R. Flamary, M. Corneli, T. Vayer, and N. Courty, "Template based graph neural network with optimal transport distances," in Advances in Neural Information Processing Systems, vol. 35, pp. 11800–11814, 2022.

Appendix

A. Warm Starting the Kernels

Upon random initialisation of the node embedding parameters and kernel coefficients, we calculate the sparse codes of all signals in the training set S and fit the kernel coefficients of each class dictionary, alternating with repeated sparse coding, to attain low reconstruction error. When the class dictionaries are of the same size, this procedure is equivalent to fitting one dictionary and copying its kernel coefficients to all class dictionaries.

As outlined in (5), the optimization procedure is done on a per-channel basis, and so let us write out the problem for channel $k \in \{1, ..., C\}$. Denote $R_x := [x(1)U\Gamma_K, ..., x(M)U\Gamma_K]$. Suppose the signal $f = [f_1^T, ..., f_C^T]^T$ yielded a sparse code r from (2) in the last sparse coding step. Rewrite (5) as such:

$$\begin{aligned}\frac{1}{2}||f_k - [r(1)U\Gamma_K, ..., r(M)U\Gamma_K]\alpha_k||_2^2 \\ = \frac{1}{2}||f_k - R_r\alpha_k||_2^2 \\ \equiv \frac{1}{2}\alpha_k^T R_r^T R_r \alpha_k - (R_r^T f_k)^T \alpha_k,\end{aligned}$$

where we dropped a constant in the last line. Now, sum this least squares reformulation over all pairs of signals f_k and their associated sparse codes r from the training set S:

$$\frac{1}{2}\alpha_k^T \left(\sum_S R_r^T R_r\right) \alpha_k - \left(\sum_S R_r^T f_k\right)^T \alpha_k.$$

Minimising this objective is done by solving the (regularized) normal equation

$$\left(\sum_S R_r^T R_r + \epsilon I_{MK}\right) \alpha_k = \sum_S R_r^T f_k$$

with the conjugate gradient descent method, where I_{MK} denotes the $MK \times MK$ identity matrix.